

Explanation of Lifecycle Methods

- **createState**: This method creates the state object for the widget. When you create a stateful widget, the framework calls `createState()` and it must be overridden.

```
class MyPage extends StatefulWidget {  
  @override  
  _MyPageState createState() =>  
  _MyPageState();  
}
```

- **initState**: This method is called after the state object is created. It is used to initialise the state of the widget. It is called once and only once.

Dart

```
@override  
void initState() {  
  super.initState();  
  print("initState");  
  _counter = 0;  
}
```

- **didChangeDependencies**: This method is called immediately after `initState` and when the dependencies of the State object change via an `InheritedWidget`.

Dart

```
@override  
void didChangeDependencies() {  
  super.didChangeDependencies();  
  print("didChangeDependencies");  
}
```

- **build**: This method is called after the state object is initialized. It is used to build the widget tree. This gets called each time the widget is rebuilt.

```
@override
Widget build(BuildContext context) {
  print("build");
  return Scaffold(
    appBar: AppBar(
      title: const Text("Lifecycle Demo"),
    ),
    body: Center(
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Text(_counter.toString()),
          ElevatedButton(
            onPressed: _increment,
            child: const Text("Increment"),
          ),
        ],
      ),
    ),
  );
}
```

- **didUpdateWidget**: This method is called when the widget is updated with new properties.

```
@override
void didUpdateWidget(covariant MyPage oldWidget) {
  super.didUpdateWidget(oldWidget);
  print("didUpdateWidget");
}
```

- **setState**: This method is called to notify the framework that data has changed. It will then re-render the widget.

```
void _increment() {
  setState(() {
    _counter++;
  });
}
```

- **deactivate**: This method is called when the State object is removed from the tree temporarily.

```
@override
void deactivate() {
  super.deactivate();
  print("deactivate");
}
```

- **Dispose**: This method is used when the State object is permanently removed from the tree. It is used to release any resources used by the widget.

Dart

```
@override
void dispose() {
  super.dispose();
  print("dispose");
}
```

This example provides a complete overview of the widget lifecycle methods and demonstrates how they work in a Flutter application.